Victor Santana

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RELEVANT SKILLS

- **Programming (in order of proficiency):** C#, C++, Python, Java, C, Rust, HTML/CSS, Node.js, Assembly
- **Embedded & Hardware:** Arduino, STM32, KiCad, Altium, LTSpice, TinkerCad, Logisim Evolution, FPGA, HLS, Verilog/SystemVerilog
- **Lab Equipment:** Oscilloscopes, Function Generators, Digital Multimeters, Soldering
- **Tools & Platforms:** Git/GitHub, VS Code, API Integration, AI-Assisted Development, OOP, Serial Communication, Networking, Linux (Ubuntu), Vitis/Vivado, Docker, Make/CMake
- **Leadership & Languages:** Professional Written & Oral Communication, Team Management, Technical Training, Mentoring, Spanish

EDUCATION

Bachelor of Science in Computer Engineering — Anticipated: July 2027
Brigham Young University - Idaho — *Rexburg, Idaho*

- **Certificates:** Computer Programming & Embedded Systems
- **GPA:** 4.0/4.0
- **Scholarship:** Brigham Young University-Idaho Grant 2024-2027 (Acknowledgement of Academic Excellence)
- **IEEE Student Branch Vice Chair:** Directed multi-disciplinary engineering teams, orchestrated hands-on technical workshops, and executed industry networking events

EXPERIENCE

Faculty Technology Center Engineering Lead — May 2024 - Present
Brigham Young University - Idaho — *Rexburg, Idaho*

- Reduced operational costs by $600 annually by architecting and deploying local CI/CD runners to computerize software deployment workflows
- Refurbished and repurposed 2 legacy storage equipment, applying hands-on hardware management to deploy reliable, zero-cost infrastructure for the Media team
- Directed and managed a 10-member engineering team, overseeing web applications, AI tools, and microcontroller projects to ensure consistent year-round delivery

Undergraduate Research Assistant (IMMERSE-X) — May 2026 - July 2026
Brigham Young University — *Provo, Utah*

- Engineered an automated, 6-script toolchain using Docker and QEMU to generate reproducible PYNQ/PetaLinux OS images for the AMD AUP-ZU3 board
- Developed custom Python I2C telemetry scripts to read raw, memory-mapped silicon registers directly from the OV5640 image sensor and Xilinx FPGA D-PHY IPs
- Utilized oscilloscopes to physically verify high-speed MIPI CSI-2 camera interfaces on the AUP-ZU3 board. Probed differential clock lanes of Pcam 5C modules to definitively validate the hardware failure hypothesis
- Provided a comprehensive technical presentation at IMMERSE 2026 detailing the Zynq UltraScale+ architecture, PYNQ image automation, and 2 MIPI camera debugging methodologies to a technical audience

PROJECTS

- **AI FPGA Vision | Team Project (August 2026):** Co-developed a real-time, hardware-accelerated QVGA video pipeline on a Xilinx Artix-7 FPGA using Verilog to capture parallel 8-bit camera streams into an asynchronous dual-port BRAM frame buffer
- **MiniSentinel.io | Personal (Jul 2026):** Designed an ESP32-based remote surveillance network utilizing C++ firmware for computerized Wi-Fi recovery and offline SD card buffering, routing live hardware telemetry and video payloads to a custom web server on a Linux environment
- **Smart RC Car | BYU STEM Camp Team Project (May 2026 - July 2026):** Engineered an ESP32-CAM Wi-Fi web server and motor control framework, featuring a WebSocket-driven mobile dashboard to provide sub-millisecond MJPEG video streaming and real-time hardware tuning for an educational workshop